

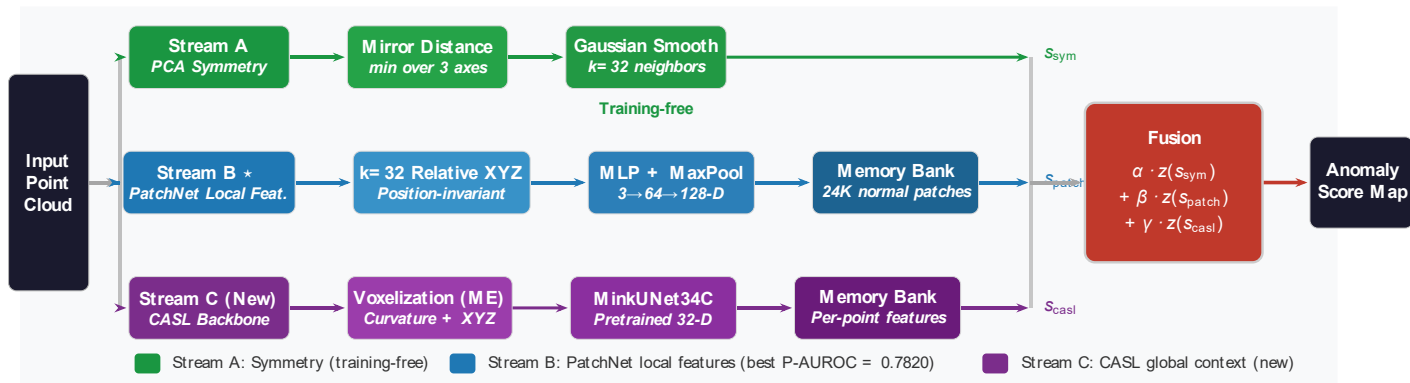


Symmetry-Aware 3D Point Cloud Anomaly Detection

12

Jun-Ting Wu, Kuan-Hung Chen

Advisor: Professor Jih-Ciang Wu



Motivation

Most industrial and natural objects possess geometric symmetry (bilateral, rotational). Anomalies — dents, bumps, missing parts — break this symmetry locally.

We exploit this as a training-free anomaly signal: no annotations, no complex training, just geometric reasoning. We combine a symmetry score with a feature-based memory bank for complementary coverage.

Methodology

- Stream A — Symmetry Score · training-free**

Mirror the PCA-aligned cloud across each axis; score = NN distance to the reflection.

$$S_{\text{sym}} = \text{GaussSmooth} \left(\min_k \min_q \|p - q\| \right)$$

→ No training; localizes asymmetric defects.

- Stream B — PatchNet Local Features · self-supervised**

Self-supervised k-NN patch encoder;

score = NN distance to memory bank.

$$S_{\text{patch}} = \min_{m \in \mathcal{M}_B} \|f_B - m\|$$

→ Position-invariant; best single stream

- Stream C — CASL Backbone Global Features · new**

Pretrained MinkUNet34C global features;

score = NN distance to memory bank.

$$S_{\text{casl}} = \min_{m \in \mathcal{M}_C} \|f_C - m\|$$

→ Adds global shape context the other two lack.

Experimental Results

Method	helmet0	bag0	bottle0	bottle1
Reg3D-AD	60.0	70.6	48.6	69.5
IMRNet	59.7	66.0	55.2	70.0
R3D-AD	75.7	72.0	73.3	73.7
PO3AD	76.2	85.7	93.8	97.1
CASL	78.3	94.8	95.7	95.4
Sym3D (Ours)	83.4	95.7	96.1	95.2

Table 1. Object-level Detection Performance (AUC-ROC %) on Anomaly-ShapeNet Dataset

Method	helmet0	bag0	bottle0	bottle1
Reg3D-AD	60.0	71.5	88.6	69.6
IMRNet	59.8	66.8	55.6	70.2
ISMP	68.3	74.7	77.0	56.8
PO3AD	87.8	98.3	94.1	93.2
CASL	92.3	96.8	85.3	86.2
Sym3D (Ours)	91.2	93.1	95.0	94.3

Table 2. Point-level Detection Performance (AUC-ROC %) on Anomaly-ShapeNet Dataset

Our Ideas

Symmetry as a training-free prior — We turn geometric symmetry into a direct anomaly signal: no labels, no training, just geometric reasoning.

Three-stream complementary fusion — We fuse symmetry, local patch features (PatchNet), and global context (CASL backbone) with learnable weights, covering both local and global anomalies.

Reference

- [1] Ye, J., et al. PO3AD: Predicting Point Offsets toward Better 3D Point Cloud Anomaly Detection. CVPR 2025
- [2] Zha, Y., et al. CASL: Curvature-augmented self-supervised learning for 3d anomaly detection. AAAI 2026

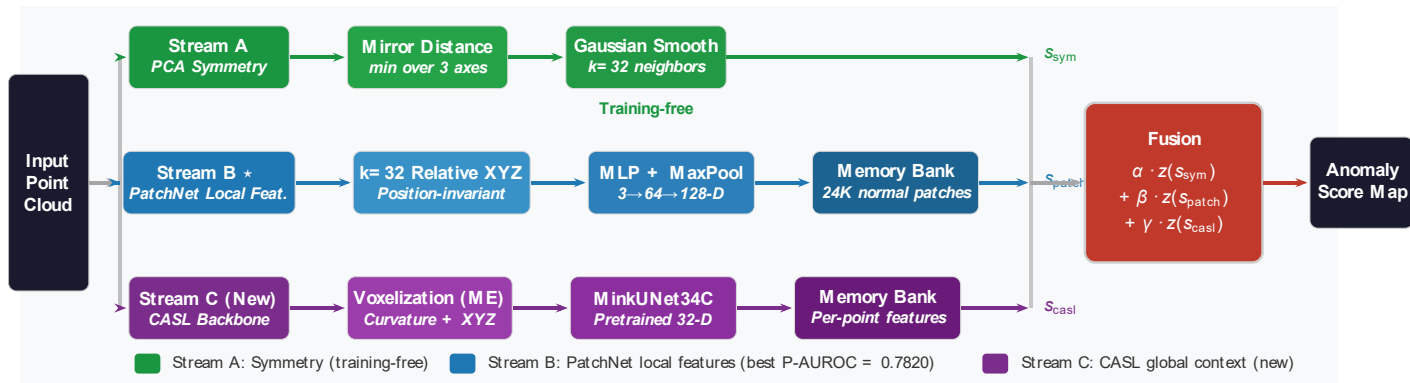


Symmetry-Aware 3D Point Cloud Anomaly Detection

12

Jun-Ting Wu, Kuan-Hung Chen

Advisor: Professor Jih-Ciang Wu



Motivation

Most industrial and natural objects possess geometric symmetry (bilateral, rotational). Anomalies — dents, bumps, missing parts — break this symmetry locally.

We exploit this as a training-free anomaly signal: no annotations, no complex training, just geometric reasoning. We combine a symmetry score with a feature-based memory bank for complementary coverage.

Methodology

- Stream B — PatchNet Local Features · self-supervised

Self-supervised k-NN patch encoder;

score = NN distance to memory bank.

$$s_{\text{patch}} = \min_{m \in \mathcal{M}_B} \|f_B - m\|$$

→ Position-invariant; best single stream (P-AUROC 0.782).

Experimental Results

Method	ashtray0	bag0	bottle0	bottle1
Reg3D-AD	59.7	70.6	48.6	69.5
IMRNet	67.1	66	55.2	70
R3D-AD	83.3	72	73.3	73.7
PO3AD	100	85.7	93.8	97.1
Sym3D				

Table 1. Object-level Detection Performance (AUC-ROC %) on Anomaly-ShapeNet Dataset

Method	ashtray0	bag0	bottle0	bottle1
Reg3D-AD	69.8	71.5	88.6	69.6
IMRNet	67.1	66.8	55.6	70.2
ISMP	60.3	74.7	77	56.8
PO3AD (Our Reproduce)	94	98.3	94.1	93.2

Table 2. Point-level Detection Performance (AUC-ROC %) on Anomaly-ShapeNet Dataset

Our Ideas

Extend symmetry prior to partial / rotationally symmetric objects (e.g., cylinders, spheres where one axis is dominant).

Combine symmetry score with PO3AD-style offset prediction for finer localization.

Explore learned symmetry axes for objects without obvious bilateral symmetry.

Reference

- [1] Ye, J., et al. PO3AD: Predicting Point Offsets toward Better 3D Point Cloud Anomaly Detection. CVPR 2025
- [2] Zha, Y., et al. CASL: Curvature-augmented self-supervised learning for 3d anomaly detection. AAAI 2026

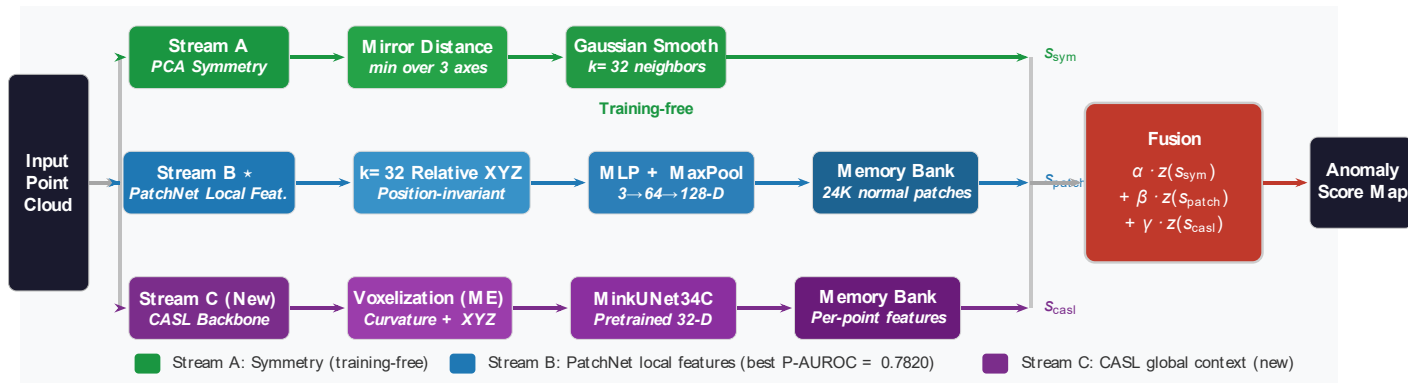


Symmetry-Aware 3D Point Cloud Anomaly Detection

12

Jun-Ting Wu, Kuan-Hung Chen

Advisor: Professor Jihh-Ciang Wu



Motivation

Most industrial and natural objects possess geometric symmetry (bilateral, rotational). Anomalies — dents, bumps, missing parts — break this symmetry locally.

We exploit this as a training-free anomaly signal: no annotations, no complex training, just geometric reasoning. We combine a symmetry score with a feature-based memory bank for complementary coverage.

Methodology

- Stream C — CASL Backbone Global Features · new Pretrained MinkUNet34C global features; score = NN distance to memory bank.

$$s_{casl} = \min_{m \in \mathcal{M}_c} \|f_c - m\|$$

→ Adds global shape context the other two lack.

Experimental Results

Method	ashtray0	bag0	bottle0	bottle1
Reg3D-AD	59.7	70.6	48.6	69.5
IMRNet	67.1	66	55.2	70
R3D-AD	83.3	72	73.3	73.7
PO3AD	100	85.7	93.8	97.1
Sym3D				

Table 1. Object-level Detection Performance (AUC-ROC %) on Anomaly-ShapeNet Dataset

Method	ashtray0	bag0	bottle0	bottle1
Reg3D-AD	69.8	71.5	88.6	69.6
IMRNet	67.1	66.8	55.6	70.2
ISMP	60.3	74.7	77	56.8
PO3AD (Our Reproduce)	94	98.3	94.1	93.2

Table 2. Point-level Detection Performance (AUC-ROC %) on Anomaly-ShapeNet Dataset

Our Ideas

Extend symmetry prior to partial / rotationally symmetric objects (e.g., cylinders, spheres where one axis is dominant).

Combine symmetry score with PO3AD-style offset prediction for finer localization.

Explore learned symmetry axes for objects without obvious bilateral symmetry.

Reference

- [1] Ye, J., et al. PO3AD: Predicting Point Offsets toward Better 3D Point Cloud Anomaly Detection. CVPR 2025
- [2] Zha, Y., et al. CASL: Curvature-augmented self-supervised learning for 3d anomaly detection. AAAI 2026